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Central authority for certifying unbonded access to a vehicle via a digital key device

Abstract

Techniques for certifying unbonded access to a vehicle are disclosed herein. An example method includes receiving, by a vehicle and from a computing system, a service identifier associated with a function of the vehicle, wherein the vehicle stores one or more device identifiers for each digital key device in communication with the vehicle, receiving, from a digital key device, a signed token associated with the service identifier, wherein the one or more device identifiers do not include a device identifier for the digital key device, sending, to the computing system, the signed token, receiving, from the computing system, an indication of whether the signed token is verified, responsive to determining, based on the indication, that the signed token is verified, authorizing use of the function of the vehicle, and storing the device identifier for the digital key device.

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Background/Summary

BACKGROUND

(1) A vehicle may have one or more digital keys that may be used to, for example, unlock and operate the vehicle. A digital key may be linked with the vehicle's owner, such as through ownership records of a vehicle manufacturer assigning the digital key to an account or other record

of the vehicle's owner, to prevent unauthorized access to or use of the vehicle. When multiple digital keys are issued for the vehicle, each digital key may be similarly associated with the vehicle's owner for these reasons.

SUMMARY

(2) In general, aspects of the techniques of this disclosure are directed to a central authority for certifying unbonded access to a vehicle via a digital key device. For example, autonomous vehicles, such as autonomous taxis, or taxi of ride share services may verify users (e.g., passengers), such that the user entering a vehicle may be confirmed as the correct user (e.g., the user that requested transport via the vehicle).

(3) Some solutions may allow digital keys to be shared with multiple users when the users complete a registration process that links these users with the owner of the vehicle. For example, individual digital keys along with an account of the user to which the digital key was issued may be stored, such as in an ownership record or database of the vehicle's manufacturer. To obtain such an account, the user may be required to complete a registration process, including one or more verification and/or authorization steps. In some examples, the vehicle's owner, the user, or both must verify their identities and authorize issuance of the digital key to the user before the user can be linked to the vehicle owner.

(4) In accordance with the techniques disclosed herein, rather than requiring users to be linked (e.g., bonded) with a vehicle owner, the central authority may certify both users and vehicles at their time of interaction, such as at the start of their itinerary (e.g., trip, journey, route), during their itinerary, or both. As such, the central authority certifies unbonded access to a vehicle allowing the user to enter, operate vehicle functions, or both. As will be described further below, the unbonded access may verify both users and vehicles, which may provide safety and convenience in use cases, such as autonomous vehicles or ride sharing services, where one vehicle may have any number of unrelated users or where a vehicle owner wishes to share their vehicle.

(5) In some aspects, the techniques described herein relate to a method including receiving, by a vehicle and from a computing system, a service identifier associated with a function of the vehicle, wherein the vehicle stores one or more device identifiers for each digital key device in communication with the vehicle; after advertising, by the vehicle, the service identifier, receiving, by the vehicle and from a digital key device, a signed token associated with the service identifier, wherein the one or more device identifiers do not include a device identifier for the digital key device; sending, by the vehicle and to the computing system, the signed token; receiving, by the vehicle and from the computing system, an indication of whether the signed token is verified; responsive to determining, by the vehicle and based on the indication, that the signed token is verified, authorizing, by the vehicle, use of the function of the vehicle; and storing, by the vehicle, the device identifier for the digital key device.

(6) In some aspects, the techniques described herein relate to a computing device including a memory that stores instructions; and one or more processors that execute the instructions to: receive, from a computing system, a service identifier associated with a function of the vehicle, wherein the vehicle stores one or more device identifiers for each digital key device in communication with the vehicle; after advertising the service identifier, receive, from a digital key device, a signed token associated with the service identifier, wherein the one or more device identifiers do not include a device identifier for the digital key device; send, to the computing system, the signed token; receive, from the computing system, an indication of whether the signed token is verified; responsive to determining, based on the indication, that the signed token is verified, authorize use of the function of the vehicle; and store the device identifier for the digital key device.

(7) In some aspects, the techniques described herein relate to a non-transitory computer-readable storage medium including instructions, that when executed by one or more processors of a computing device, cause the one or more processors to: receive, from a computing system, a

service identifier associated with a function of the vehicle, wherein the vehicle stores one or more device identifiers for each digital key device in communication with the vehicle; after advertising the service identifier, receive, from a digital key device, a signed token associated with the service identifier, wherein the one or more device identifiers do not include a device identifier for the digital key device; send, to the computing system, the signed token; receive, from the computing system, an indication of whether the signed token is verified; responsive to determining, based on the indication, that the signed token is verified, authorize use of the function of the vehicle; and store the device identifier for the digital key device.

(8) The details of one or more examples of the subject matter of this specification are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages of the subject matter will become apparent from the description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a conceptual diagram illustrating an example environment for certifying unbonded access to a vehicle via a digital key device, in accordance with one or more aspects of the present disclosure.

(2) FIG. 2 is a block diagram illustrating an example vehicle, in accordance with one or more aspects of the present disclosure.

(3) FIG. 3 is a flowchart of an example process for certifying unbonded access to a vehicle via a digital key device, in accordance with one or more aspects of the present disclosure.

(4) FIG. 4 is a conceptual diagram illustrating an example process for certifying unbonded access to a vehicle via a digital key device, in accordance with one or more aspects of the present disclosure.

DETAILED DESCRIPTION

(5) FIG. 1 is a conceptual diagram illustrating an example environment for certifying unbonded access to a vehicle via a digital key device, in accordance with one or more aspects of the present disclosure. As can be seen, environment **100** may include a computing device **110**, a computing system **130**, and a vehicle **140**. As described herein, computing system **130** may constitute a central authority for certifying unbonded access to vehicle **140**, and computing device **110** may be referred to as a digital key device **110** in that computing device/digital key device **110** may be used to obtain access to vehicle **140** and vehicle functions.

(6) Computing device **110** may be an example of a mobile phone, a tablet computer, a laptop computer, a wearable device, a gaming system, a media player, an e-book reader, or any other type of computing device that may operate as a digital key. FIG. 1 illustrates a particular example of computing device **110**, and many other examples of computing device **110** may be used in other instances and may include a subset of the components included in example computing device **110** or may include additional components not shown in FIG. 1.

(7) Computing device **110** includes user interface device **112**, one or more processors **102**, one or more input devices **104**, one or more wireless communication units **108**, one or more output devices **106**, and one or more memory devices **114**. One or more memory devices **114** of computing device **110** may include operating system **116**, client module **118**, and data store **122**. Data store **122** may store various data, such as in a structured or unstructured format. For example, data store **122** may be a database or a file folder for storing various data. For instance, data store **122** may store one or more service identifiers **124**, one or more signed tokens **126**, one or more device identifiers **128**, other data, or various subsets thereof.

(8) Communication channels **148** may interconnect each of the components **102**, **104**, **106**, **108**, and **112** for inter-component communications (physically, communicatively, and/or operatively). In

some examples, communication channels **148** may include a system bus, a network connection, an inter-process communication data structure, or any other method for communicating data.

(9) One or more input devices **104** of computing device **110** may receive input. Examples of input are tactile, audio, and video input. Input devices **104** of computing device **110**, in one example, includes a presence-sensitive display, touch-sensitive screen, mouse, keyboard, voice responsive system, video camera, microphone or any other type of device for detecting input from a human or machine.

(10) One or more output devices **106** of computing device **110** may generate output. Examples of output are tactile, audio, and video output. Output devices **106** of computing device **110**, in one example, includes a presence-sensitive display, sound card, video graphics adapter card, speaker, liquid crystal display (LCD), organic light-emitting diode (OLED) display, a light field display, haptic motors, linear actuating devices, or any other type of device for generating output to a human or machine.

(11) One or more wireless communication units **108** of computing device **110** may communicate with external devices by transmitting and/or receiving communication signals, such as via one or more wireless networks or wireless connections. Examples of one or more wireless communication units **108** include a network interface card, an optical transceiver, a radio frequency transceiver, a global positioning system (GPS) receiver, or any other type of device that can wirelessly send and/or receive information. Other examples of one or more wireless communication units **108** may include short wave radios, cellular data radios, wireless network radios, as well as universal serial bus (USB) controllers. Though described with respect to wireless communication units, computing device **110** may also or alternatively include one or more wired communication units, such as network interface cards (e.g., Ethernet cards), optical transceivers, or any other type of device that can send and/or receive information over a wired connection.

(12) User interface device **112** of computing device **110** may be hardware that functions as an input and/or output device for computing device **110**. For example, user interface device **112** may include a display component, which may be a screen at which information is displayed by user interface device **112** and a presence-sensitive input device that may detect an object at and/or near the display component.

(13) One or more processors **102** may implement functionality and/or execute instructions within computing device **110**. For example, one or more processors **102** on computing device **110** may receive and execute instructions stored by one or more memory devices **114** that execute the functionality of operating system **116** and client module **118**. The instructions executed by one or more processors **102** may cause computing device **110** to store information within one or more memory devices **114** during program execution. Examples of one or more processors **102** include application processors, display controllers, sensor hubs, and any other hardware configured to function as a processing unit. One or more processors **102** may execute instructions of operating system **116** and client module **118** to perform actions or functions. That is, operating system **116** and client module **118** may be operable by one or more processors **116** to perform various actions or functions of computing device **110**.

(14) One or more memory devices **114** within computing device **110** may store information for processing during operation of computing device **110**. That is, computing device **110** may store data accessed by operating system **116** and client module **118** during execution at computing device **110**, including service identifiers **124**, stored tokens **126**, device identifiers **128**, other data, or various subsets thereof. In some examples, memory device **114** is a temporary memory, meaning that a primary purpose of memory device **114** is not long-term storage. One or more memory devices **114** on computing device **110** may be configured for short-term storage of information as volatile memory and therefore not retain stored contents if powered off. Examples of volatile memories include random access memories (RAM), dynamic random access memories (DRAM), static random access memories (SRAM), and other forms of volatile memories known in the art.

(15) One or more memory devices **114**, in some examples, also include one or more computer-readable storage media. One or more memory devices **114** may be configured to store larger amounts of information than volatile memory. One or more memory devices **114** may further be configured for long-term storage of information as non-volatile memory space and retain information after power on/off cycles. Examples of non-volatile memories include magnetic hard discs, optical discs, floppy discs, flash memories, or forms of electrically programmable memories (EPROM) or electrically erasable and programmable (EEPROM) memories. One or more memory devices **114** may store program instructions and/or information (e.g., data) associated with operating system **116** and client module **118**.

(16) Client module **118**, may execute at one or more processors **102** to perform functions related to certifying unbonded access. In some examples, client module **118** may operate in conjunction with computing system **130**. For example, client module **118** may be used to create a user or vehicle account at computing system **130** for a user of computing device **110** or vehicle **140**, respectively. Account information, such as authentication information (e.g., username, password, multi-factor authentication tokens), user information (e.g., name and address information), payment information (e.g., credit card or bank account information), user preferences (e.g., account settings), subscription/membership information, or various subset thereof, may be received by client module **118**. For example, client module **118** may receive account information from a user via user interface device **112**, input device **104**, or both such as to sign up the user. Client module **118** may transmit the account information to computing system **130** via wireless communication unit **108** and network **144**. Computing system **130** may process and/or store the account information to create an account. The account may be required before client module **118** may transmit or receive service identifiers **124**, signed tokens **126**, or both, such as to or from computing system **130**.

(17) Client module **118** may be used to create one or more itineraries, such as for a trip, journey, route, or task. For example, client module **118** may receive itinerary information including a start time and/or end time (e.g., a start and/or end date), a starting location and ending location (e.g., starting/ending coordinates or addresses), a route (e.g., driving directions), a task (e.g., parcel/food delivery), or other calendaring or scheduling information. For instance, client module **118** may receive itinerary information indicating a route and start time from the user via user interface device **112**, input device **104**, or both. Client module **118** may transmit the itinerary information to computing system **130** and computing system **130** may process and/or store the itinerary information to create an itinerary for the user. Client module **118** may receive confirmation of the itinerary from computing system **130** that confirms that the itinerary has been created.

(18) In some examples, client module **118** may transmit a start notification identifying the itinerary to computing system **130**, such as via wireless communication unit **108**. The notification may indicate that client module **118** is ready to start the itinerary. As will be described further below, responsive to receipt of the start notification, computing system **130** may generate and transmit one or more service identifiers **124** and one or more signed tokens **126** used to certify unbonded access to a vehicle **140**.

(19) The start notification may signify to computing system **130** that client module **118** is ready to begin the itinerary (e.g., start a trip, journey, or route). The start notification may be transmitted for various reasons. For example, client module **118** may transmit the start notification at the start time of the itinerary or a predefined period of time before the start time of the itinerary. In some examples, client module **118** may transmit the start notification based on location or proximity in some examples. For instance, client module **118** may transmit the start notification upon determining computing device **110** is within a predefined distance (e.g., 10 ft) of a starting location of the itinerary. In some examples, client module **118** may determine computing device **110** is within a predefined distance of the starting location using a GPS receiver and GPS signals, such as by determining that the GPS coordinates of computing device **110** are within the predefined distance of the GPS coordinates of the starting location.

(20) In some examples, client module **118** may transmit the start notification to computing system **130** upon determining vehicle **140** and computing device **110** are within a predefined distance (e.g., 10 ft). For instance, client module **118** may determine computing device **110** and vehicle **140** are within a predefined distance using a GPS receiver and GPS signals, such as by determining that the GPS coordinates of computing device **110** and vehicle **140** are within the predefined distance. Client module **118** may determine computing device **110** and vehicle **140** are within a predefined distance without GPS, such as by computing device **110** receiving and/or measuring the strength of one or more signals from vehicle **140**, or vice versa, such as via wireless communication unit **108**.

(21) Network **144** represents any public or private communications network, for instance, cellular, Wi-Fi, and/or other types of networks, for transmitting data between computing systems, servers, and computing devices. Network **144** may include one or more network hubs, network switches, network routers, or any other network equipment, that are operatively inter-coupled thereby providing for the exchange of information between computing system **130** and computing device **110**. Computing device **110**, vehicle **140**, and computing system **130** may transmit and receive data across network **144** using any suitable communication techniques. For example, account information, itinerary information, service identifiers **124**, and signed tokens **126** may be transmitted and received between computing system **130** and computing device **110** as well as computing system **130** and vehicle **140** via network **144**. Each of computing device **110**, vehicle **140**, and computing system **130** may be operatively coupled to network **144** using respective network links, such as Ethernet, Wi-Fi, or any other types of wired and/or wireless network connections.

(22) Vehicle **140** may be an example of a car, truck, boat, aircraft, train, bicycle, motorcycle, scooter, skateboard, or any other type of motorized or non-motorized vehicle. FIG. **1** illustrates a particular example of vehicle **140**, and many other examples of vehicle **140** may be used in other instances and may include a subset of the components included in example vehicle **140** or may include additional components not shown in FIG. **1**. Vehicle **140** may include one or more device implements **142** or components that provide various functions or features of vehicle **140**. Example device implements **142** may include devices such as actuators, motors, and electronic systems. In some examples, device implements **142** may include door locks, ignition systems, steering/control systems, transmission systems (e.g., automatic or manual transmissions), infotainment systems, door/window actuation systems (e.g., door/window motors). Unbonded access to vehicle **140** may include unbonded access to device implements **142** of vehicle **140** in some examples. For example, unbonded access to vehicle **140** may permit a user to unlock and enter vehicle **140**, drive or operate a vehicle **140**, and/or use other functions of vehicle **140** implemented via device implements **142**.

(23) As shown in FIG. **1**, vehicle **140** may include computing device **120**. In some examples, computing device **120** may be a computer or computing device installed in vehicle **140**, such as an electronic control unit (ECU), navigation system, or infotainment system. Computing device **110** and computing device **120** of vehicle **140** may perform similar operations when certifying unbonded access to vehicle **140**. As such, in some examples, computing device **120** may be an example computing device **110**, except that, computing device **120** may not be considered a digital key device at least for the reason that computing device **120** is integral to (e.g., a component of) vehicle **140**.

(24) In accordance with the techniques disclosed herein, computing device **120** and computing device **110** may operate in conjunction with computing system **130** to certify unbonded access to vehicle **140**. For example, computing system **130**, with computing device **120** and computing device **110**, may certify unbonded access to vehicle **140**, without requiring computing device **120**/vehicle **140** and computing device **110** to be previously paired or linked.

(25) FIG. **2** is a block diagram illustrating an example vehicle, in accordance with one or more aspects of the present disclosure. As can be seen, vehicle **240** may include one or more computing devices **220** and one or more device implements **242**. In FIG. **2**, vehicle **240**, device implements

242, and computing device 220 may respectively be examples of vehicle 140, device implements 142, and computing device 120 of FIG. 1. FIG. 2 illustrates a particular example of computing device 220, and many other examples of computing device 220 may be used in other instances and may include a subset of the components included in example computing device 220 or may include additional components not shown in FIG. 2.

(26) Similar to computing device 110 of FIG. 1, computing device 210 includes user interface device 212, one or more processors 202, one or more input devices 204, one or more wireless communication units 208, one or more output devices 206, and one or more memory devices 214. One or more memory devices 214 of computing device 210 may include operating system 216, client module 218, and data store 222. Data store 222 may store various data, such as in a structured or unstructured format. For example, data store 222 may be a database or a file folder. In some examples, data store 122 may store one or more service identifiers 224, one or more device identifiers 228, or both.

(27) As described above, computing device 110 and computing device 220 may perform similar operations when certifying unbonded access to vehicle 240. As such, processor 202, input device 204, output device 206, wireless communication unit 108, user interface device 212, memory device 214, operating system 216, client module 218, and data store 222, communication channels 248 of computing device 220 FIG. 2 may respectively be examples of processor 102, input device 104, output device 106, wireless communication unit 108, user interface device 112, memory device 114, operating system 116, client module 118, and data store 122, communication channels 148 of computing device 110 FIG. 1. Similarly, service identifier 224, signed token 226, and device identifier 228 of FIG. 2 may respectively be examples of service identifier 124, signed token 126, and device identifier 128 of FIG. 1.

(28) Referring to FIGS. 1-2, computing system 130 may be any suitable remote computing system, such as one or more desktop computers, laptop computers, mainframes, servers, cloud computing systems, virtual machines, etc. capable of sending and receiving information via network 144. In some examples, computing system 130 may represent a cloud computing system that provides one or more services via network 144. That is, in some examples, computing system 130 may be a distributed computing system. One or more computing devices, such as computing device 110, may access the services provided by the cloud by communicating with computing system 130.

(29) In some examples, computing system 130 includes one or more processors, one or more wireless communication devices, and one or more memory devices. A memory device of computing system 130 may include an operating system, which may provide an execution environment for a certification module 132 that, when executed by one or more processors, certifies unbonded access to vehicle 140 as described herein. A memory device of computing system 130 may include a data store 134 that stores various data, such as in a structured or unstructured format. For example, data store 134 may be a database or a file folder. In some examples, data store 122 may store account information 136, itinerary information 138, or other data.

(30) Computing system 130, computing device 110, and computing device 120 of vehicle 140 may operate in conjunction to certify unbonded access to vehicle 140. For example, as described above, computing system 130 may receive a start notification from computing device 110. In response, computing system 130 may generate service identifier 124 and signed token 126 and transmit service identifier 124 and signed token 126 to computing device 110, such as via network 144. Similarly, computing system 130 may receive a start notification from computing device 120 of vehicle 140. In response, computing system 130 may generate service identifier 224 and signed token 226 and transmit service identifier 224 and signed token 226 to computing device 120, such as via network 144.

(31) Service identifier 124, 224 may be various identifiers in some examples. For instance, computing system 130 may generate service identifier 124, 224 by generating an alphanumeric

character string or binary data with a random number generator or by execution of a heuristic for generating identifiers (e.g., a universally unique identifier generator). In some examples, computing system **130** may encrypt service identifier **124**, **224**. Signed token **126**, **226** may be a cryptographic or otherwise encrypted or encoded token. For example, signed token **126**, **226** may be cryptographically signed such that computing system **130** may verify that computing system **130** generated signed token **126**, **226**. Computing system **130** may sign or encrypt signed token **126**, **226** such as through various encryption methods, including public key infrastructure (PKI) encryption methods. Computing system **130** may store a copy of service identifiers **124**, **224**, signed tokens **126**, **226**, or both such as in data store **134**.

(32) As shown in the example of FIG. **1**, computing device **110** may receive, from computing system **130**, service identifier **124** and signed token **126** and store service identifier **124** and signed token **126** on memory device **114**. Computing device **120** may receive, from computing system **130**, service identifier **224** and signed token **226** and store service identifier **224** and signed token **226** on memory device **214**. Computing devices **110**, **120** may advertise (e.g., transmit) their respective service identifiers **124**, **224**. Computing devices **110**, **120** may transmit service identifiers **124**, **224** wireless communication units **108**, **208** using various wireless protocols. For example, computing devices **110**, **120** may transmit service identifiers **124**, **224** as a generic attribute profile (GATT) characteristic or identifier, such as via a wireless communication unit **108**, **208** comprising a BLUETOOTH low energy (BLE) radio.

(33) Computing devices **110**, **120** may utilize service identifiers **124**, **224** to establish a connection or communication link **146** therebetween. For example, computing device **120** may advertise (e.g., transmit) service identifier **124** and computing device **110** may scan for (e.g., receive) service identifiers. Computing device **110** may compare service identifier **124** to service identifier **224** and determine whether service identifier **124** corresponds to or matches service identifier **224**. If service identifier **124** and service identifier **224** correspond or match, computing device **110** may establish a communication link **146** with computing device **120**, such as shown in FIG. **1**. The communication link **148** may be wireless and may be established by computing devices **110**, **120** via respective wireless communication units **108**, **208**. In some examples, advertising and scanning may be reversed. For instance, computing device **110** may advertise service identifier **224** and computing device **120** of vehicle **140** may scan for service identifiers. In such an example, computing device **120** may compare service identifier **224** with service identifier **124** and establish a communication link **146** when service identifier **224** and service identifier **124** match.

(34) Communication link **146** may be various communication links. In some examples, computing devices **110**, **120** may establish a secure (e.g., encrypted) or insecure (e.g., unencrypted) communication link **146**. For example, computing device **110**, **120** may establish an insecure logical link control and adaptation protocol (L2CAP) connection via respective wireless communication units **108**, **208**. Though described as a wireless communication link **146**, communication link **146** may be wired in some examples. For example, vehicle **140** and computing device **110** may be in electrical/optical communication via one or more electrical/optical conductors or connectors when certification of unbonded access occurs.

(35) Computing devices **110**, **120** may exchange or share signed tokens **126**, **226** with one another. For example, computing device **110** may transmit signed token **126** to computing device **120** and computing device **120** may transmit signed token **226** to computing device **110**. Computing device **110**, **120** may verify signed tokens **126**, **226** received from the other computing device **110**, **120**. For example, computing device **110** may verify signed token **226** and computing device **120** may verify signed token **126**.

(36) Computing devices **110**, **120** may verify signed tokens **126**, **226** by transmitting signed tokens **126**, **226** to computing system **130** for verification. For example, computing device **110** may transmit signed token **226** (received from computing device **120**) to computing system **130** verification and computing device **120** may transmit signed token **126** (received from computing

device **110**) to computing system **130** for verification. Computing system **130** may verify signed token **126**, **226** in various ways. In some examples, computing system **130** may verify signed token **126**, **226** by verifying a signature of signed token **126**, **226**. If necessary, computing system **130** may decrypt signed token **126**, **226**, such as with a private key of computing system **130** where signed token **126**, **226** was encrypted with a public key of computing system **130**.

(37) Computing system **130** may indicate whether signed token **126**, **226** was verified. For example, responsive to receiving signed token **226** from computing device **110**, computing system **130** may transmit a verification notification to computing device **110** indicating whether or not signed token **226** was verified. Similarly, responsive to receiving signed token **126** from computing device **120**, computing system **130** may transmit a verification notification to computing device **120** indicating whether or not signed token **126** was verified.

(38) Computing device **110**, **120** may perform one or more functions or refrain from the same based on whether or not their respective signed token **126**, **226** was verified by computing system **130**. For example, responsive to receiving a verification notification that signed token **126** was verified, computing device **120** may authorize use of a function of vehicle **140**, such as be authorizing use of device implement **142** of vehicle **140**. For instance, computing device **120** may unlock a door of vehicle **140** or permit the ignition of vehicle **140** to be started. With respect to computing device **110**, responsive to receiving a verification notification that signed token **226** was verified, computing device **110** may present a message indicating that vehicle **140** is the vehicle **140** requested or referenced by the itinerary created by the user of computing device **110**. For example, computing device **110** may present a message, such as via user interface device **112** or output device **106**, indicating the vehicle **140** was verified and is safe to enter.

(39) FIG. 3 is a flowchart of an example process for certifying unbonded access to a vehicle via a digital key device, in accordance with one or more aspects of the present disclosure. FIG. 3 is described in the context of FIGS. 1-2.

(40) In some examples, vehicle **240** may store one or more device identifiers **228** for each digital key device **110** in communication with or that has established communication with vehicle **240**. Vehicle **240** may receive from computing system **130**, service identifier **224** associated with a function of vehicle **240** (**302**). For example, computing system **130** may store service identifier **224** in data store **134** along with an itinerary indicating a function of vehicle **240** to be used during itinerary (e.g., unlock door or drive vehicle). Some example functions of vehicle **240** include unlocking one or more doors, starting the vehicle, selecting a drive mode (e.g., gear), and driving the vehicle. Vehicle **240** may advertise service identifier **224**, such as via wireless communication unit **208**.

(41) After advertising service identifier **224**, vehicle **240** may receive signed token **126** associated with service identifier **224** (**304**). For example, digital key device **110** may store service identifier **224** and scan for service identifiers, such as service identifier **224**. Upon determining a service identifier, such as advertised service identifier **224** corresponds to or matches service identifier **124**, digital key device **110** may transmit signed token **126** to vehicle **240**. Vehicle **240** may then receive signed token **126**. In this example, device identifiers **228** currently stored by vehicle **240** may not include device identifier **128** for digital key device **110**, indicating that vehicle **240** and digital key device **110** have not previously established communication or communicated.

(42) Vehicle **240** may send signed token **126** to computing system **130** (**306**). Vehicle **240** may receive an indication of whether signed token **126** is verified from computing system **130** (**308**). Responsive to determining, by vehicle that signed token **126** is verified based on the indication from computing system **130**, vehicle **240** may authorize use of the function of vehicle **240** (**310**). Vehicle **240** may store device identifier **128** for digital key device **110** (**312**), such as device identifier **228** of memory device **214**. When vehicle **240** determines, based on the indication, that signed token **126** is not verified, vehicle **240** may refrain from authorizing use of the function of vehicle **240**. In some examples, vehicle **240** may send an indication that vehicle **240** is disabled

(e.g., damaged, out of fuel) to computing system **130**. Computing system **130** may respond by refraining from verifying signed token **126** (e.g., transmitting an indication that signed token **126** is not verified) when vehicle **240** is disabled.

(43) Digital key device **110** may verify signed token **226** of vehicle **240** in some examples. For instance, digital key device **110** may receive signed token **226** from vehicle **240** and send signed token **226** to computing system **130** for verification. Digital key device **110** may receive, from computing system **130**, an indication signed token **226** was verified by computing system **130**. In response to receiving the indication of the verification of signed token **226**, digital key device **110** may perform a predefined function. For example, digital key device **110** may present a notification that vehicle **240** is verified to convey to the user of digital key device **110** that the vehicle **240** is the requested vehicle. Vehicle **240** may receive signed token **226** from computing system **130** and digital key device **110** may receive signed token **126** from computing system **130** to allow vehicle **240** and computing device **110** to exchange signed tokens **126**, **226** for verification as described above.

(44) In some examples, vehicle **240** may receive, from computing system **130**, an indication of an itinerary (e.g., itinerary information) including an occurring time period (e.g., a start and end time) within which the itinerary is to occur. When vehicle **240** determines a current time is outside the occurring time period, vehicle **240** may refrain from authorizing use of the function of vehicle **240**.

(45) Vehicle **240** may establish a communication link **146** with digital key device **110** for communication purposes in some examples. Vehicle **240** may receive one or more commands from digital key device **110** via the communication link **146**. The commands may control operation of the function of vehicle **240**. For example, the commands may control door locks, infotainment, or navigation systems of vehicle **240**. Vehicle **240** may terminate communication link **146** based on one or more criteria received from computing system **130**. For example, vehicle **240** may terminate communication at the end of an itinerary, such as at an ending time of the itinerary or when an ending location of the itinerary is reached.

(46) FIG. **4** is a conceptual diagram illustrating an example process for certifying unbonded access to a vehicle via a digital key device, in accordance with one or more aspects of the present disclosure. FIG. **4** is described below in the context of FIGS. **1-2**.

(47) In some examples, a user **460** of digital key device **110** and vehicle **240** may have accounts at computing system **130**. For instance, user **460**, such as via digital key device **110** or other computing device, may create an account by sending the user's account information to computing system **130** (**402**) and vehicle **240** may create an account by an operator/user of vehicle **240** sending the vehicle's account information to computing system **130** (**404**). Computing system **130** may receive the account information of both user **460** and vehicle **240** and store such account information (**406**), such as in data store **134**.

(48) User **460**, such as via digital key device **110** or other computing device, may create an itinerary at computing system **130** (**408**). Once the itinerary is created, computing system **130** may share (e.g., transmit) the itinerary with digital key device **110** and vehicle **240** (**410**). At a start of the itinerary, digital key device **110** and vehicle **240** may respectively transmit a start notification to computing system **130** (**412**, **414**). For example, digital key device **110** and vehicle **240** may transmit a start notification at a starting time of the itinerary or when digital key device **110** and vehicle **240** are within a predetermined distance of one another.

(49) In response, computing system **130** may generate signed tokens **126**, **226** (**416**) and send signed tokens **126**, **226** to vehicle **240** and digital key device **110**. For example, computing system **130** may send signed token **226** to vehicle **240** and vehicle **240** may receive signed token **226**. Computing system **130** may send signed token **126** to digital key device **110** and digital key device **110** may receive signed token **126**. Digital key device **110** may store signed token **126** (**418**) and vehicle **240** may store signed token **226** (**420**).

(50) Computing system **130** may generate service identifiers **124**, **224** (**422**) and send service

identifiers **124, 224** to digital key device **110** and vehicle **240**. In some examples, computing system **130** may generate service identifiers **124, 224** in response to receiving the start notification. Vehicle **240** may advertise service identifier **224 (424)** and digital key device **110** may scan for service identifier **224 (426)**. Upon detecting (e.g., receiving) service identifier **224**, digital key device **110** may compare service identifier **224** to service identifier **124** and determine if service identifiers **124, 224** correspond to or match one another.

(51) Responsive to determining a correspondence or match between service identifiers **124, 224**, digital key device **110** and vehicle **240** may establish communication link **146** between digital key device **110** and vehicle **240 (428, 430)**. Digital key device **110** and vehicle **240** may utilize communication link **146** to exchange signed tokens **126, 226**. For example, digital key device **110** may transmit signed token **126** to vehicle **240 (432)** and vehicle **240** may transmit signed token **226** to digital key device **110 (434)**.

(52) Digital key device **110** and vehicle **240** may subsequently verify exchanged signed tokens **126, 226**. For example, digital key device **110** may transmit signed token **226** received from vehicle **240** to computing system **130** and vehicle **240** may transmit signed token **126** received from digital key device **110** to computing system **130**. Computing system **130** may verify signed tokens **126, 226 (436)**. For example, computing system **130** may verify signed token **126** received from vehicle **240** and signed token **226** received from digital key device **110**.

(53) Computing system **130** may transmit an indication of whether signed token **126, 226** is verified or not verified. For example, computing system **130** may transmit an indication of whether signed token **126** is verified or not verified to vehicle **240** and transmit an indication of whether signed token **226** is verified or not verified to digital key device **110**. When vehicle **240** receives an indication that signed token **126** is verified by computing system **130**, vehicle **240** may authorize use of a function of vehicle **240 (438)**. For example, vehicle **240** may unlock or permit an ignition system to be started. When digital key device **110** receives an indication that signed token **226** is verified by computing system **130**, digital key device **110** may perform a predefined function **(440)**. For example, digital key device **110** may present a message, such as via user interface device **112**, confirming that vehicle **240** is the requested vehicle **240**.

(54) Though described with respect to vehicles, the techniques disclosed herein may be used with various devices. For example, rather than a vehicle, unbonded access may be certified for mailboxes, delivery lockers, and other devices. For example, a delivery locker may include an example of computing device **120** and operate in conjunction with computing system **130** and digital key device **110** as described herein to allow the delivery locker to authorize access to the contents of delivery lock (e.g., unlock delivery locker via certified unbonded access).

(55) Example 1: A method includes receiving, by a vehicle and from a computing system, a service identifier associated with a function of the vehicle, wherein the vehicle stores one or more device identifiers for each digital key device in communication with the vehicle; after advertising, by the vehicle, the service identifier, receiving, by the vehicle and from a digital key device, a signed token associated with the service identifier, wherein the one or more device identifiers do not include a device identifier for the digital key device; sending, by the vehicle and to the computing system, the signed token; receiving, by the vehicle and from the computing system, an indication of whether the signed token is verified; responsive to determining, by the vehicle and based on the indication, that the signed token is verified, authorizing, by the vehicle, use of the function of the vehicle; and storing, by the vehicle, the device identifier for the digital key device.

(56) Example 2: The method of example 1, further includes responsive to determining, by the vehicle and based on the indication, that the signed token is not verified, refraining, by the vehicle, from authorizing use of the function of the vehicle.

(57) Example 3: The method of example 1, further includes establishing, by the vehicle, a communication link with the digital key device; and receiving, by the vehicle, one or more commands from the digital key device via the communication link.

(58) Example 4: The method of example 3, wherein the one or more commands control operation of the function of the vehicle.

(59) Example 5: The method of example 3, further comprising terminating, by the vehicle, the communication link based on one or more criteria received from the computing system.

(60) Example 6: The method of example 1, further comprising sending, by the vehicle, an indication that the vehicle is disabled, wherein the computing system refrains from verifying the signed token when the vehicle is disabled.

(61) Example 7: The method of example 1, wherein the signed token is a first signed token, the method further includes receiving, by the digital key device, a second signed token from the vehicle; sending, by the digital key device, the second signed token to the computing system for verification; receiving, by the digital key device and from the computing system, an indication the second signed token was verified by the computing system; and responsive to receiving the indication of the verification of the second signed token, presenting, by the digital key device, a notification that the vehicle is verified.

(62) Example 8: The method of example 7, further includes receiving, by the vehicle and from the computing system, the first signed token; and receiving, by the digital key device and from the computing system, the second signed token.

(63) Example 9: The method of example 1, further comprising receiving, by the vehicle and from the computing system, an indication of an itinerary including an occurring time period within which the itinerary is to occur.

(64) Example 10 The method of example 9, further includes determining, by the vehicle, a current time is outside the occurring time period; and responsive to determining the current time is outside the occurring time period, refraining, by the vehicle, from authorizing use of the function of the vehicle:

(65) Example 11: The method of example 1, wherein the function is an action performed by the vehicle selected from the group consisting of unlocking one or more doors, starting the vehicle, and selecting a drive mode of the vehicle.

(66) Example 12: A vehicle includes a memory that stores instructions; and one or more processors that execute the instructions to: receive, from a computing system, a service identifier associated with a function of the vehicle, wherein the vehicle stores one or more device identifiers for each digital key device in communication with the vehicle; after advertising the service identifier, receive, from a digital key device, a signed token associated with the service identifier, wherein the one or more device identifiers do not include a device identifier for the digital key device; send, to the computing system, the signed token; receive, from the computing system, an indication of whether the signed token is verified; responsive to determining, based on the indication, that the signed token is verified, authorize use of the function of the vehicle; and store the device identifier for the digital key device.

(67) Example 13: The vehicle of example 12, wherein the one or more processors further execute the instructions to: responsive to determining, based on the indication, that the signed token is not verified, refrain from authorizing use of the function of the vehicle.

(68) Example 14: The vehicle of example 12, wherein the one or more processors further execute the instructions to: establish a communication link with the digital key device; and receive one or more commands from the digital key device via the communication link.

(69) Example 15: The vehicle of example 14, wherein the one or more commands control operation of the function of the vehicle.

(70) Example 16: The vehicle of example 14, wherein the one or more processors further execute the instructions to terminate the communication link based on one or more criteria received from the computing system.

(71) Example 17: The vehicle of example 12, wherein the one or more processors further execute the instructions to send an indication that the vehicle is disabled, wherein the computing system

refrains from verifying the signed token when the vehicle is disabled.

(72) Example 18: The vehicle of example 12, wherein the one or more processors further execute the instructions to receive, from the computing system, an indication of an itinerary including an occurring time period within which the itinerary is to occur.

(73) Example 19: The vehicle of example 18, wherein the one or more processors further execute the instructions to: determine a current time is outside the occurring time period; and responsive to determining the current time is outside the occurring time period, refrain from authorizing use of the function of the vehicle.

(74) Example 20: A non-transitory computer-readable storage medium including instructions, that when executed by one or more processors of a computing device, cause the one or more processors to: receive, from a computing system, a service identifier associated with a function of the vehicle, wherein the vehicle stores one or more device identifiers for each digital key device in communication with the vehicle; after advertising the service identifier, receive, from a digital key device, a signed token associated with the service identifier, wherein the one or more device identifiers do not include a device identifier for the digital key device; send, to the computing system, the signed token; receive, from the computing system, an indication of whether the signed token is verified; responsive to determining, based on the indication, that the signed token is verified, authorize use of the function of the vehicle; and store the device identifier for the digital key device.

(75) In one or more examples, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored on or transmitted over, as one or more instructions or code, a computer-readable medium and executed by a hardware-based processing unit. Computer-readable media may include computer-readable storage media, which corresponds to a tangible medium such as data storage media, or communication media including any medium that facilitates transfer of a computer program from one place to another, e.g., according to a communication protocol. In this manner, computer-readable media generally may correspond to (1) tangible computer-readable storage media, which is non-transitory or (2) a communication medium such as a signal or carrier wave. Data storage media may be any available media that can be accessed by one or more computers or one or more processors to retrieve instructions, code and/or data structures for implementation of the techniques described in this disclosure. A computer program product may include a computer-readable medium.

(76) By way of example, and not limitation, such computer-readable storage media can comprise random-access memory (RAM), read-only memory (ROM), EEPROM, compact disc read-only memory (CD-ROM) or other optical disk storage, magnetic disk storage, or other magnetic storage devices, flash memory, or any other storage medium that can be used to store desired program code in the form of instructions or data structures and that can be accessed by a computer. Also, any connection is properly termed a computer-readable medium. For example, if instructions are transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. It should be understood, however, that computer-readable storage mediums and media and data storage media do not include connections, carrier waves, signals, or other transient media, but are instead directed to non-transient, tangible storage media. Disk and disc, as used herein, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc, where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above should also be included within the scope of a computer-readable medium.

(77) Instructions may be executed by one or more processors, such as one or more digital signal

processors (DSPs), general purpose microprocessors, application specific integrated circuits (ASICs), field programmable logic arrays (FPGAs), or other equivalent integrated or discrete logic circuitry. Accordingly, the term “processor,” as used herein may refer to any of the foregoing structures or any other structures suitable for implementation of the techniques described herein. In addition, in some aspects, the functionality described herein may be provided within dedicated hardware and/or software modules. Also, the techniques could be fully implemented in one or more circuits or logic elements.

(78) The techniques of this disclosure may be implemented in a wide variety of devices or apparatuses, including a wireless handset, an integrated circuit (IC) or a set of ICs (e.g., a chip set). Various components, modules, or units are described in this disclosure to emphasize functional aspects of devices configured to perform the disclosed techniques, but do not necessarily require realization by different hardware units. Rather, as described above, various units may be combined in a hardware unit or provided by a collection of inter-operative hardware units, including one or more processors as described above, in conjunction with suitable software and/or firmware.

(79) Various embodiments have been described. These and other embodiments are within the scope of the following claims.

Claims

1. A method comprising: receiving, by a vehicle and from a computing system, a service identifier associated with a function of the vehicle, wherein the vehicle stores one or more device identifiers for each digital key device in communication with the vehicle; after advertising, by the vehicle, the service identifier, receiving, by the vehicle and from a digital key device, a signed token associated with the service identifier, wherein the one or more device identifiers do not include a device identifier for the digital key device; sending, by the vehicle and to the computing system, the signed token; receiving, by the vehicle and from the computing system, an indication of whether the signed token is verified; responsive to determining, by the vehicle and based on the indication, that the signed token is verified, authorizing, by the vehicle, use of the function of the vehicle; and storing, by the vehicle, the device identifier for the digital key device.
2. The method of claim 1, further comprising: responsive to determining, by the vehicle and based on the indication, that the signed token is not verified, refraining, by the vehicle, from authorizing use of the function of the vehicle.
3. The method of claim 1, further comprising: establishing, by the vehicle, a communication link with the digital key device; and receiving, by the vehicle, one or more commands from the digital key device via the communication link.
4. The method of claim 3, wherein the one or more commands control operation of the function of the vehicle.
5. The method of claim 3, further comprising terminating, by the vehicle, the communication link based on one or more criteria received from the computing system.
6. The method of claim 1, further comprising sending, by the vehicle, an indication that the vehicle is disabled, wherein the computing system refrains from verifying the signed token when the vehicle is disabled.
7. The method of claim 1, wherein the signed token is a first signed token, the method further comprising: receiving, by the digital key device, a second signed token from the vehicle; sending, by the digital key device, the second signed token to the computing system for verification; receiving, by the digital key device and from the computing system, an indication the second signed token was verified by the computing system; and responsive to receiving the indication of the verification of the second signed token, presenting, by the digital key device, a notification that the vehicle is verified.
8. The method of claim 7, further comprising: receiving, by the vehicle and from the computing

system, the first signed token; and receiving, by the digital key device and from the computing system, the second signed token.

9. The method of claim 1, further comprising receiving, by the vehicle and from the computing system, an indication of an itinerary including an occurring time period within which the itinerary is to occur.

10. The method of claim 9, further comprising: determining, by the vehicle, a current time is outside the occurring time period; and responsive to determining the current time is outside the occurring time period, refraining, by the vehicle, from authorizing use of the function of the vehicle.

11. The method of claim 1, wherein the function is an action performed by the vehicle selected from the group consisting of unlocking one or more doors, starting the vehicle, and selecting a drive mode of the vehicle.

12. A vehicle comprising: a memory that stores instructions; and one or more processors that execute the instructions to: receive, from a computing system, a service identifier associated with a function of the vehicle, wherein the vehicle stores one or more device identifiers for each digital key device in communication with the vehicle; after advertising the service identifier, receive, from a digital key device, a signed token associated with the service identifier, wherein the one or more device identifiers do not include a device identifier for the digital key device; send, to the computing system, the signed token; receive, from the computing system, an indication of whether the signed token is verified; responsive to determining, based on the indication, that the signed token is verified, authorize use of the function of the vehicle; and store the device identifier for the digital key device.

13. The vehicle of claim 12, wherein the one or more processors further execute the instructions to: responsive to determining, based on the indication, that the signed token is not verified, refrain from authorizing use of the function of the vehicle.

14. The vehicle of claim 12, wherein the one or more processors further execute the instructions to: establish a communication link with the digital key device; and receive one or more commands from the digital key device via the communication link.

15. The vehicle of claim 14, wherein the one or more commands control operation of the function of the vehicle.

16. The vehicle of claim 14, wherein the one or more processors further execute the instructions to terminate the communication link based on one or more criteria received from the computing system.

17. The vehicle of claim 12, wherein the one or more processors further execute the instructions to send an indication that the vehicle is disabled, wherein the computing system refrains from verifying the signed token when the vehicle is disabled.

18. The vehicle of claim 12, wherein the one or more processors further execute the instructions to receive, from the computing system, an indication of an itinerary including an occurring time period within which the itinerary is to occur.

19. The vehicle of claim 18, wherein the one or more processors further execute the instructions to: determine a current time is outside the occurring time period; and responsive to determining the current time is outside the occurring time period, refrain from authorizing use of the function of the vehicle.

20. A non-transitory computer-readable storage medium comprising instructions, that when executed by one or more processors of a computing system, cause the one or more processors to: receive, from a computing system, a service identifier associated with a function of a vehicle, wherein the vehicle stores one or more device identifiers for each digital key device in communication with the vehicle; after advertising the service identifier, receive, from a digital key device, a signed token associated with the service identifier, wherein the one or more device identifiers do not include a device identifier for the digital key device; send, to the computing

system, the signed token; receive, from the computing system, an indication of whether the signed token is verified; responsive to determining, based on the indication, that the signed token is verified, authorize use of the function of the vehicle; and store the device identifier for the digital key device.
